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Advanced Computer Networks

Network Economy and Social Impact

What happened last lecture?

IPv6 SRA, routing loops, and open DNS

IPv6 SRA + routing loops

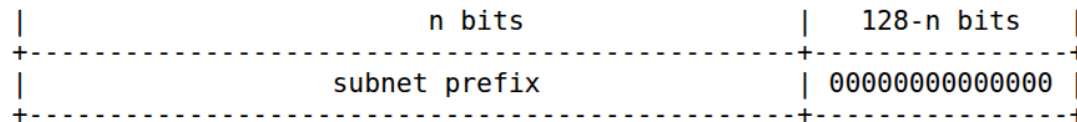
Subnet-Router Anycast (SRA) probing

A specific IPv6 address to find the closest router of a subnet

RFC 1884 (IPv6 addressing architecture):

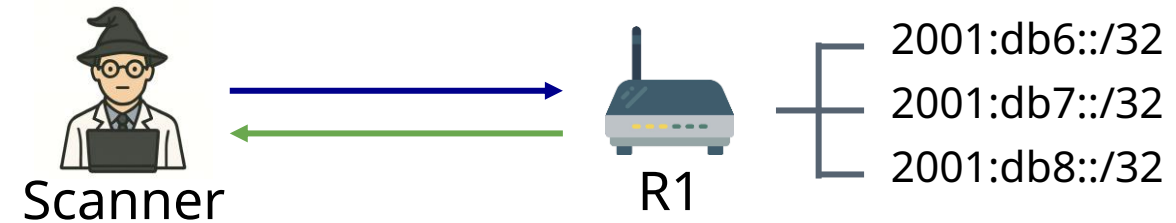
2.5.1 Required Anycast Address

The Subnet-Router anycast address is predefined. It's format is as follows:



Routers with interfaces to the corresponding subnet are **required** to reply!

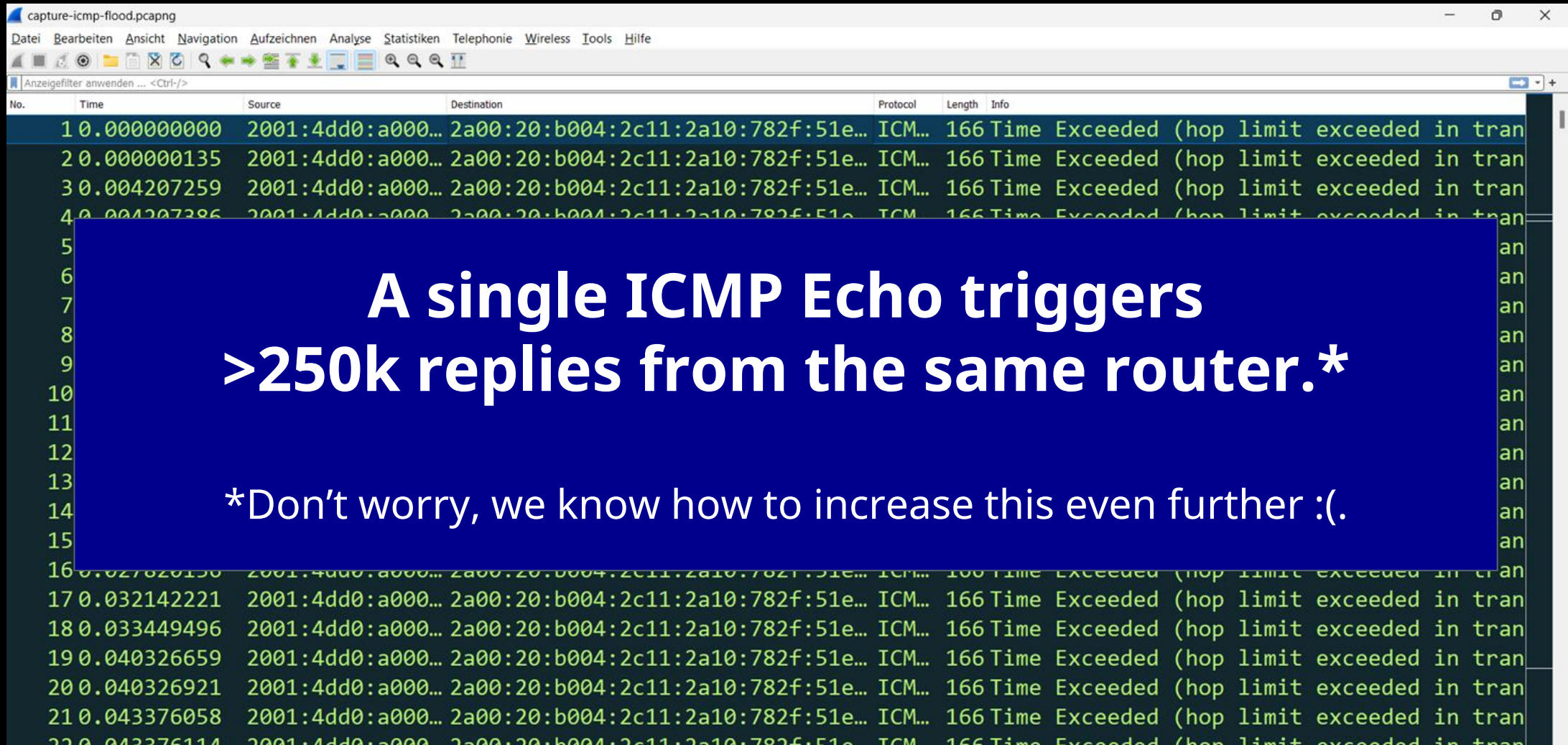
ICMP Echo Request
Dst. IP: 2001:db8::0



ICMP Echo Reply
Src. IP: R1

Helpful in case of IPv6 scanning
since it reduces ICMPv6 error messages and thus rate limiting [see Internet measurements].

```
ping -c 1 <destination IP address>
```



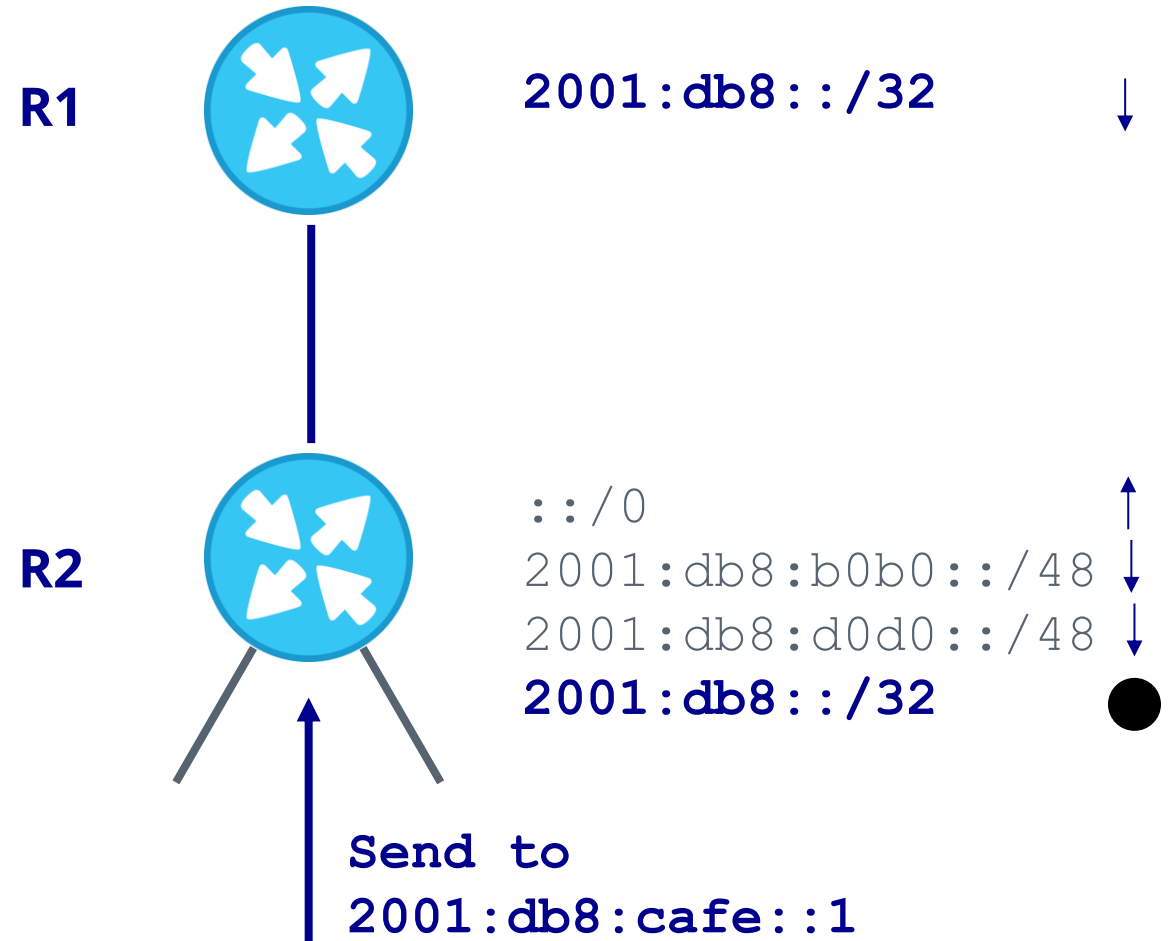
The image shows a Wireshark packet capture window titled 'capture-icmp-flood.pcapng'. The interface includes a menu bar (Datei, Bearbeiten, Ansicht, Navigation, Aufzeichnen, Analyse, Statistiken, Telefonie, Wireless, Tools, Hilfe) and a toolbar. Below the toolbar is a filter bar with the text 'Anzeigefilter anwenden ... <Ctrl-/>'. The main display area shows a list of network packets. The first four packets are visible, all being ICMP Echo (ping) requests from source 2001:4dd0:a000... to destination 2a00:20:b004:2c11:2a10:782f:51e... The 'Info' column for these packets shows 'Time Exceeded (hop limit exceeded in tran...'. A large blue text box is overlaid on the packet list, containing the text: 'A single ICMP Echo triggers >250k replies from the same router.*' and '*Don't worry, we know how to increase this even further :(.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	2001:4dd0:a000...	2a00:20:b004:2c11:2a10:782f:51e...	ICMP...	166	Time Exceeded (hop limit exceeded in tran
2	0.000000135	2001:4dd0:a000...	2a00:20:b004:2c11:2a10:782f:51e...	ICMP...	166	Time Exceeded (hop limit exceeded in tran
3	0.004207259	2001:4dd0:a000...	2a00:20:b004:2c11:2a10:782f:51e...	ICMP...	166	Time Exceeded (hop limit exceeded in tran
4	0.004207386	2001:4dd0:a000...	2a00:20:b004:2c11:2a10:782f:51e...	ICMP...	166	Time Exceeded (hop limit exceeded in tran
5						an
6						an
7						an
8						an
9						an
10						an
11						an
12						an
13						an
14						an
15						an
16	0.027820150	2001:4dd0:a000...	2a00:20:b004:2c11:2a10:782f:51e...	ICMP...	166	Time Exceeded (hop limit exceeded in tran
17	0.032142221	2001:4dd0:a000...	2a00:20:b004:2c11:2a10:782f:51e...	ICMP...	166	Time Exceeded (hop limit exceeded in tran
18	0.033449496	2001:4dd0:a000...	2a00:20:b004:2c11:2a10:782f:51e...	ICMP...	166	Time Exceeded (hop limit exceeded in tran
19	0.040326659	2001:4dd0:a000...	2a00:20:b004:2c11:2a10:782f:51e...	ICMP...	166	Time Exceeded (hop limit exceeded in tran
20	0.040326921	2001:4dd0:a000...	2a00:20:b004:2c11:2a10:782f:51e...	ICMP...	166	Time Exceeded (hop limit exceeded in tran
21	0.043376058	2001:4dd0:a000...	2a00:20:b004:2c11:2a10:782f:51e...	ICMP...	166	Time Exceeded (hop limit exceeded in tran
22	0.043376114	2001:4dd0:a000...	2a00:20:b004:2c11:2a10:782f:51e...	ICMP...	166	Time Exceeded (hop limit exceeded in tran

Why?



Preventing routing loops: Null route.



When you operate DNS resolver infrastructure, you have two options



Public

Any host on the Internet should be able to send a request and get a reply.

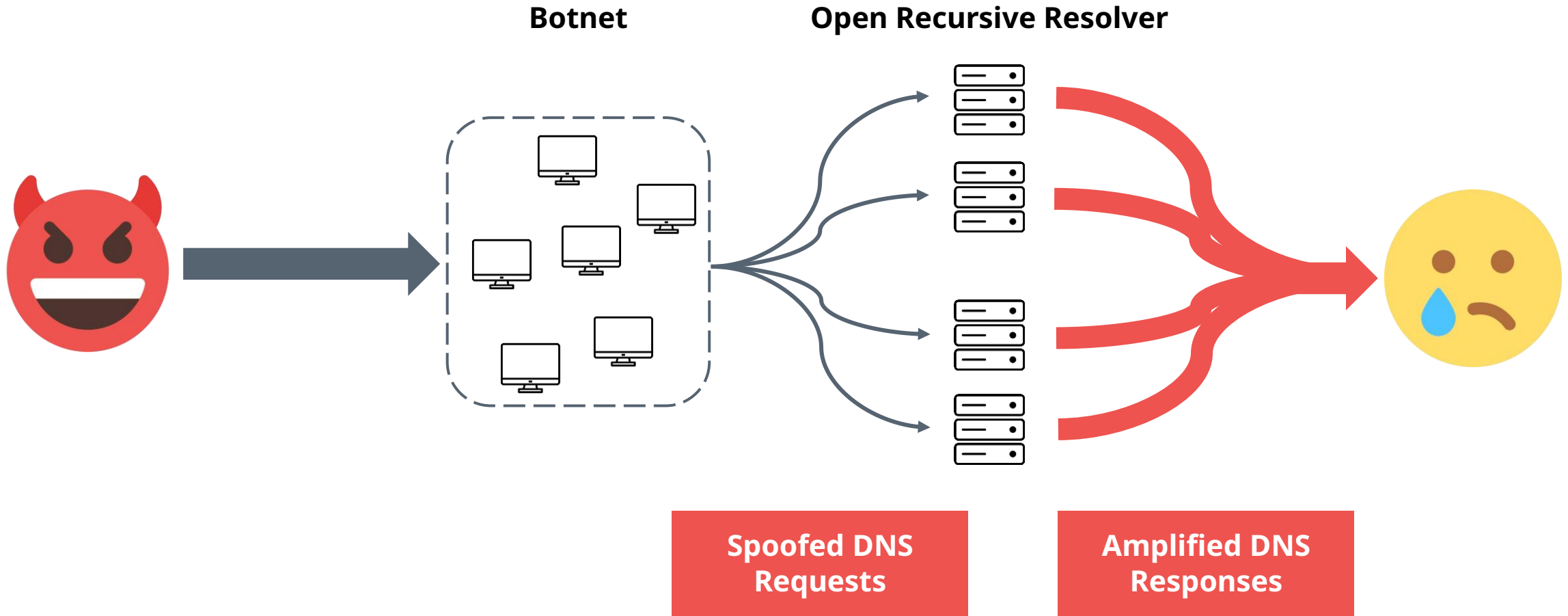


Close

Only hosts of your network should be able to send a request and get a reply.

Potential Misuse Scenario of Open Recursive Resolvers

To run a distributed attack, the attacker needs access to a botnet.



Potential Misuse Scenario of Open Recursive Resolvers

To run a distributed attack, the attacker needs access to a botnet.

Botnet

Open Recursive Resolver

Botnets are costly.

Any alternatives?

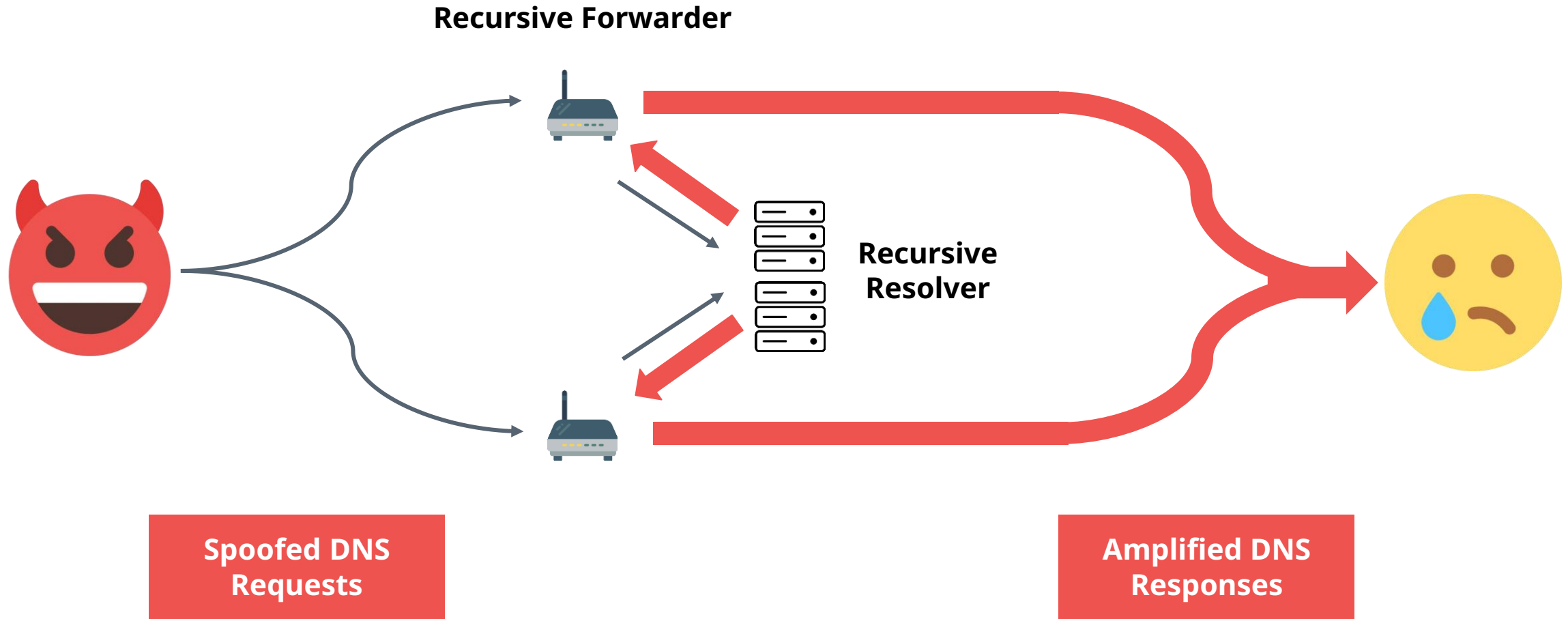


Spoofed DNS
Requests

Amplified DNS
Responses

Potential Misuse Scenario of Open Recursive Forwarders

Forwarders are globally distributed, enabling distributed attacks w/o a botnet.



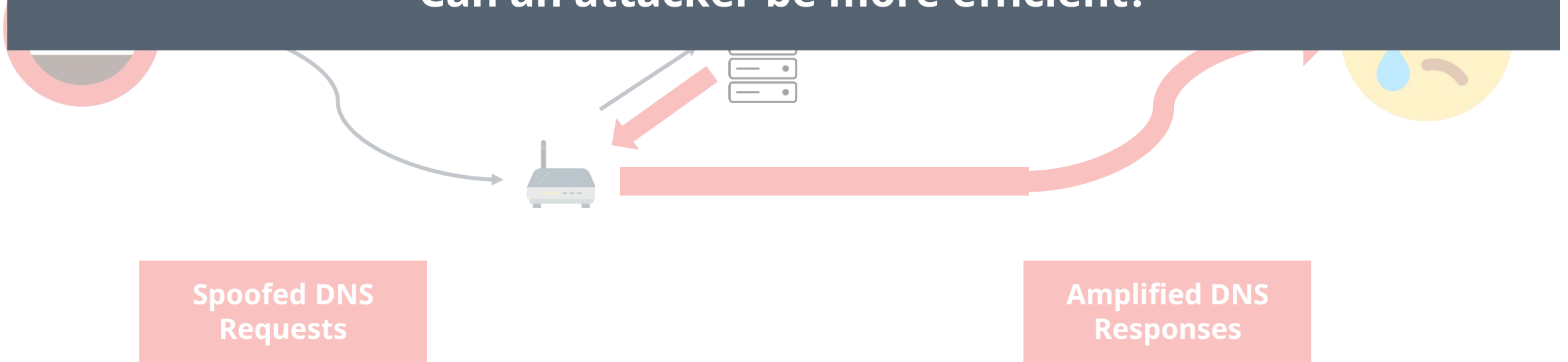
Potential Misuse Scenario of Open Recursive Forwarders

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Recursive Forwarder

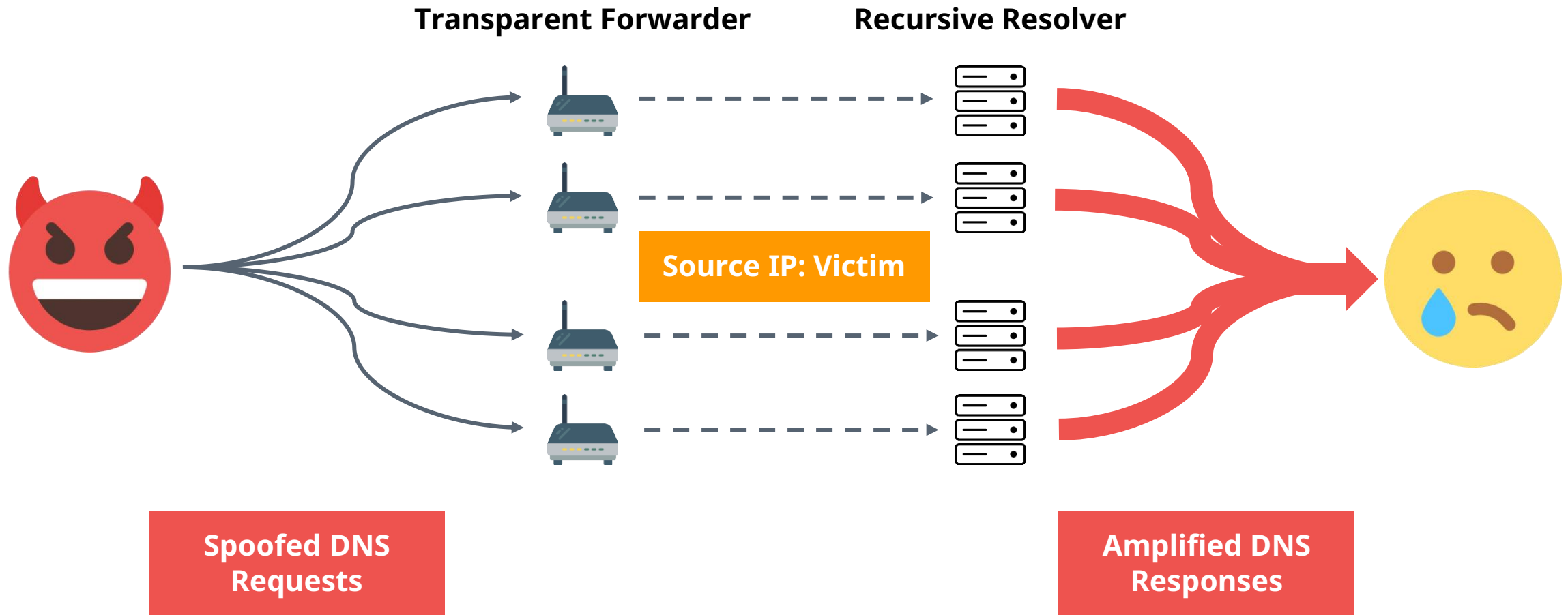
A recursive forwarder has to handle the amplified response.

Can an attacker be more efficient?



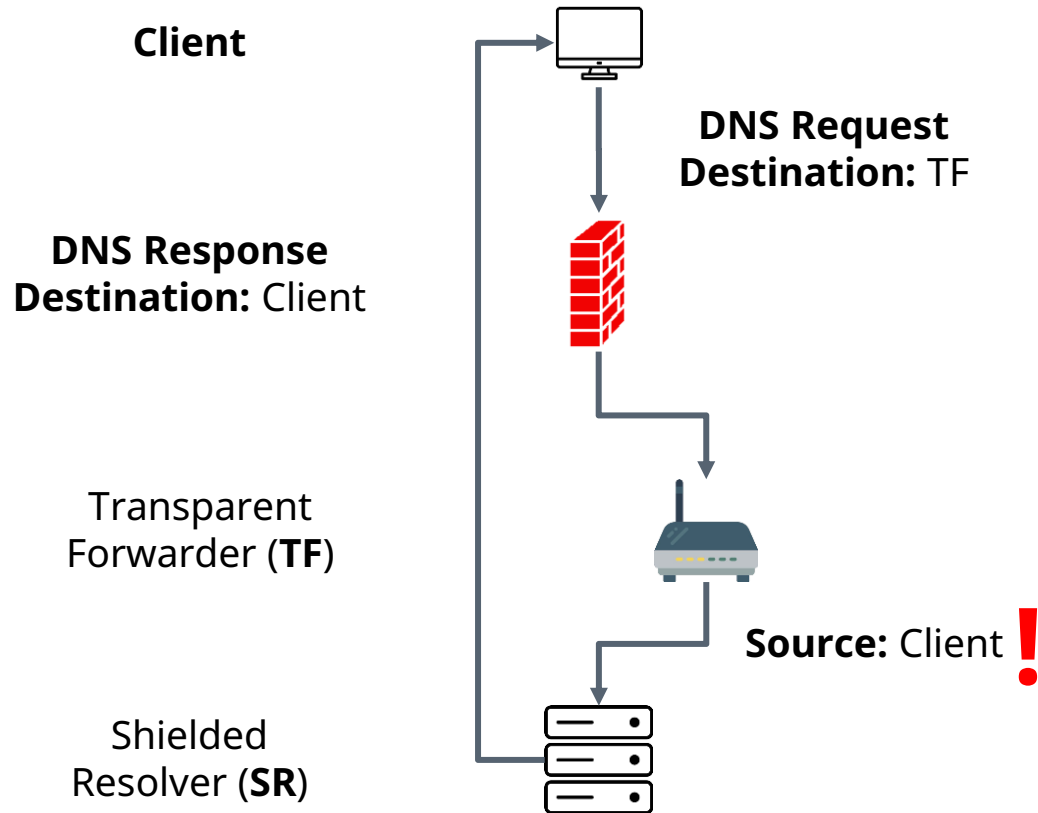
Transparent DNS Forwarders

The recursive resolver allegedly receives a DNS request from the DDoS target.



Additional Attack Infrastructure – Shielded Resolvers

Transparent forwarders enable access to shielded resolvers.



Shielded resolvers should only be accessible from the internal network.

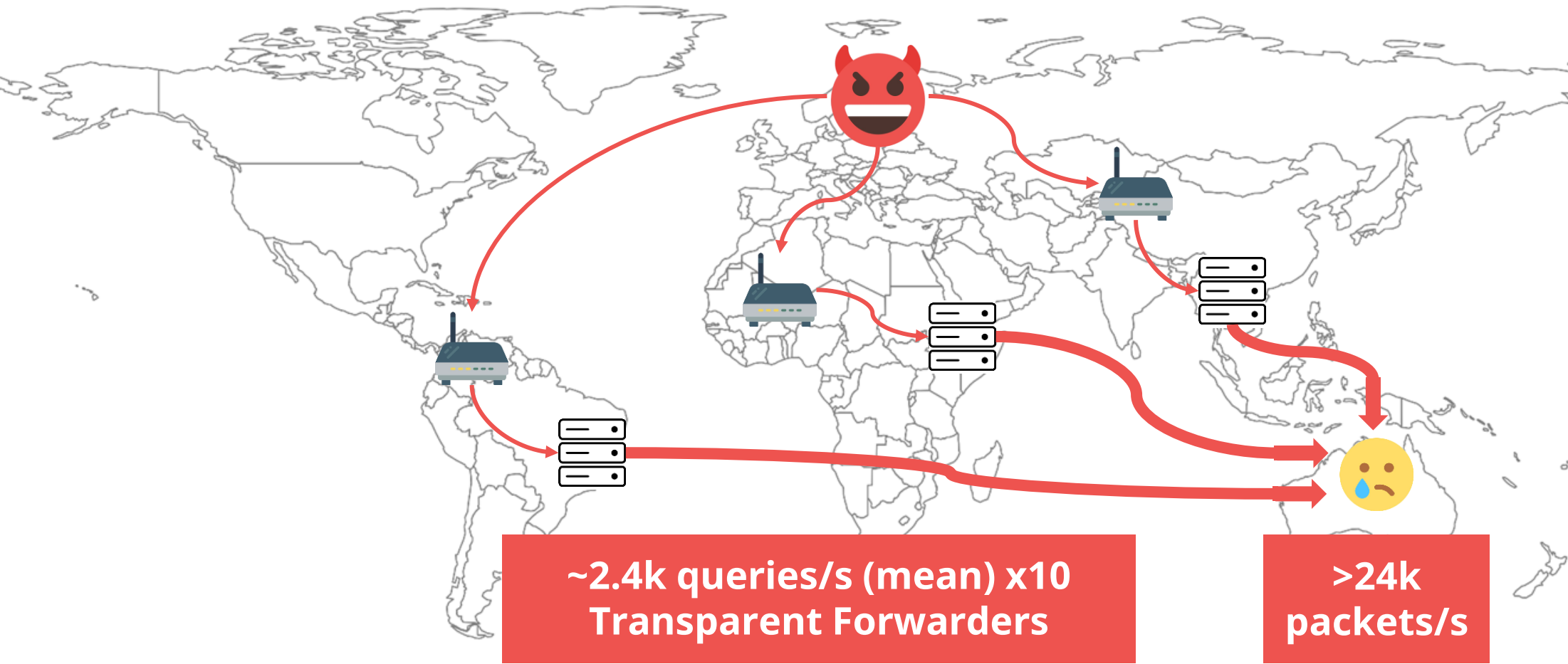
Firewall at the network border filters requests that directly target the shielded resolver.

The transparent forwarder is not affected by the firewall rules at the network border.

No properly configured firewall at the resolver, hence the client receives a valid DNS response.

Transparent Forwarders can be Utilized to Circumvent Rate Limiting

Using Internet-wide measurements, we can show an amplification factor of up to 14 by orchestrating 10 transparent forwarders.



Recent topic: Iran outage

Iran Is Cut Off From Internet as Protests Calling for Regime Change Intensify

As protests swelled around the country, Iran's internet was shut down, and the heads of its judiciary and its security services warned of a harsh response amid calls for "freedom, freedom."



Listen to this article · 7:53 min [Learn more](#)



Share full article



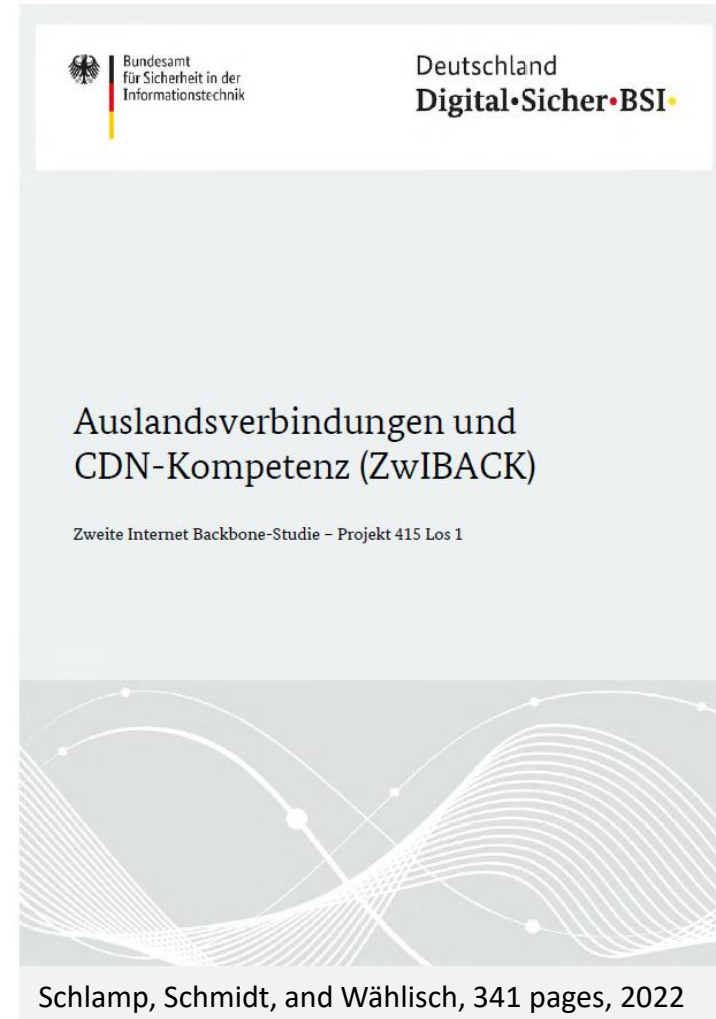
334

Iran Plunges Into Internet Blackout as Protests Intensify

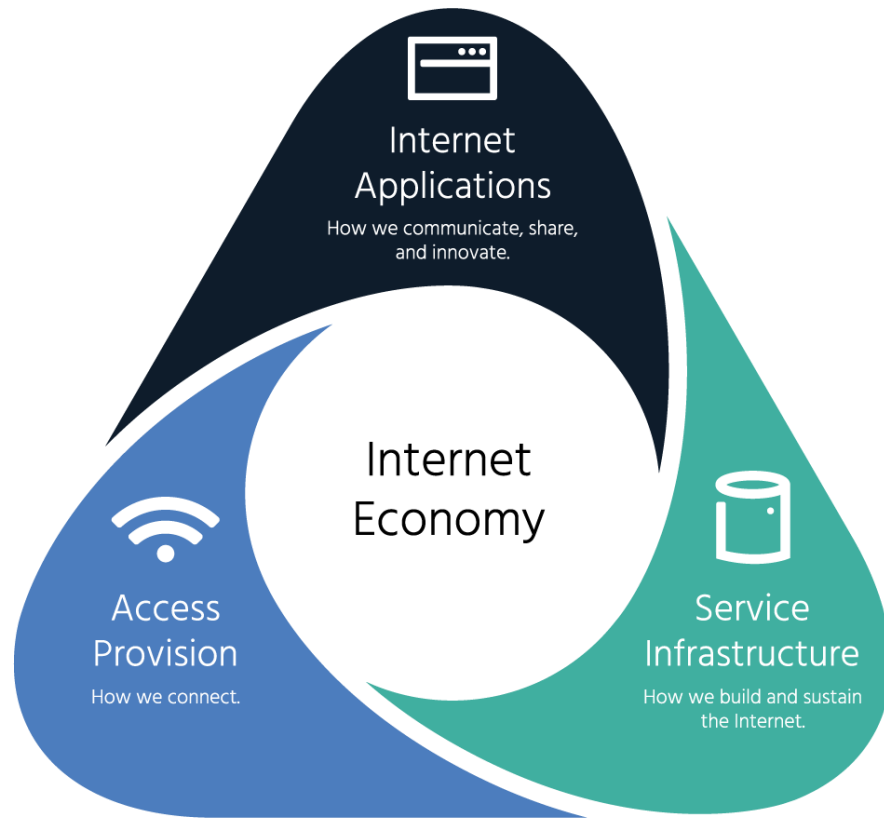
What happens this lecture?

Network Economy and Social Impact

Major parts are based on the following report



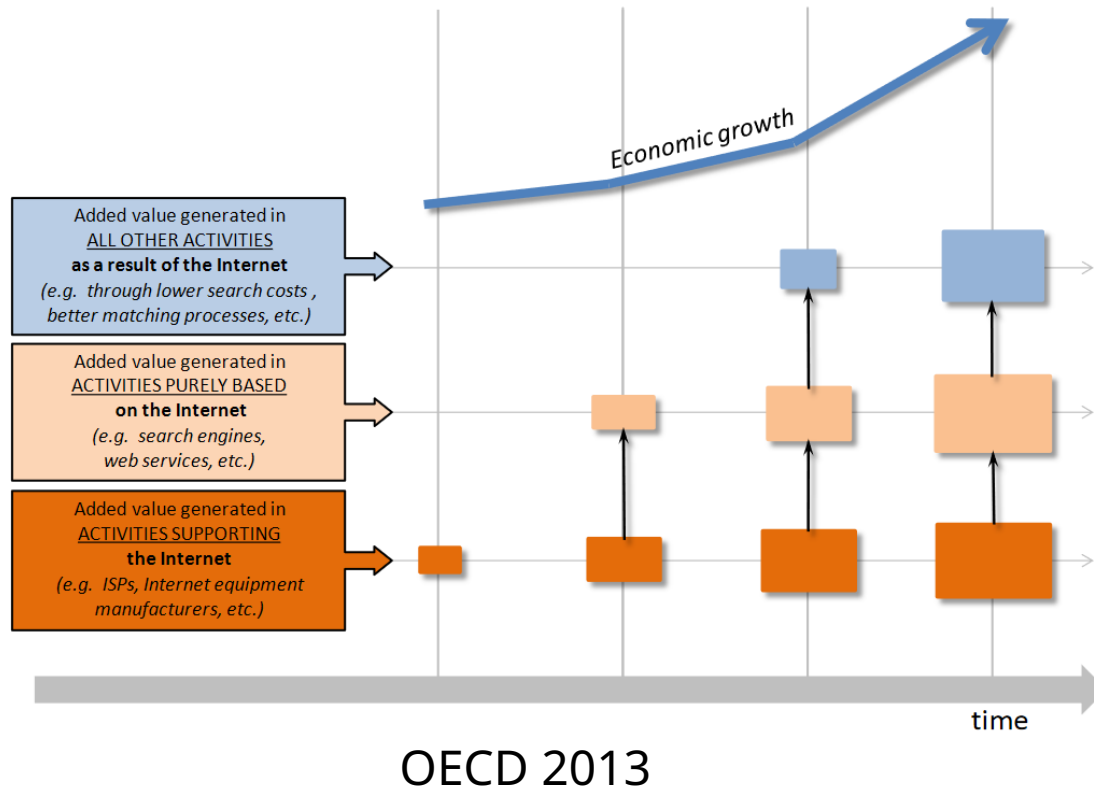
Internet economy



Economic activities that either support the Internet or are fundamentally dependent on the Internet existence

Internet Society 2019

Contributions to the Growth Domestic Product



Indirect effects s.a.

- information density
- operational synergies
- efficiency gains
- online marketing

start to dominate

An Economic Theory of Communication

A theory of interdependent demand for a communications service

Jeffrey Rohlfs

Bell Laboratories
Murray Hill, New Jersey

Rohlfs' seminal paper observed 1974:

“Larger is better for communication systems”

The utility that a subscriber derives from a communications service increases as others join the system. This is a classic case of external economies in consumption and has fundamental importance for the economic analysis of the communications industry. This paper analyzes the economic theory of this kind of interdependent demand. We begin by defining “equilibrium user set” as a set of users consistent with all individuals’ (users and nonusers) maximizing their utilities. There are typically multiple equilibria at any

The Bell Journal of Economics and Management Science, vol. 5, no. 1, pp. 16–37, 1974.

Consequences

Highly scalable communication systems

- either follow open standards
- or tend to become monopolies

As of today, closed applications have created huge “winner-takes-all” markets that stimulated consolidation even for areas, in which open standards dominate(d).

Consequences



Like the oil barons at the turn of the 20th century, the data barons are determined to extract as much as possible of a resource that's central to the economy of their time. The more information they can get to feed the algorithms that power their ad-targeting machines and product-recommendation engines, the better.

—MIT Technology Review, 2018

Highly scalable communication systems

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»EIN FREUND IST
EINER, DER ALLES
VON DIR WEISS,
UND DER DICH
TROTZDEM LIEBT.«

MARK ZUCKERBERG

----- SAMSTAG -----

04. FEBRUAR

Service infrastructure

Tier 1 and Transit

Two key services

1. Global connectivity at very high speeds
2. Global prefix reachability (DFZ)

Operationally related fields

- Transnational/subsea cable infrastructure
- Access provisioning

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- Access provisioning

Main business competitors

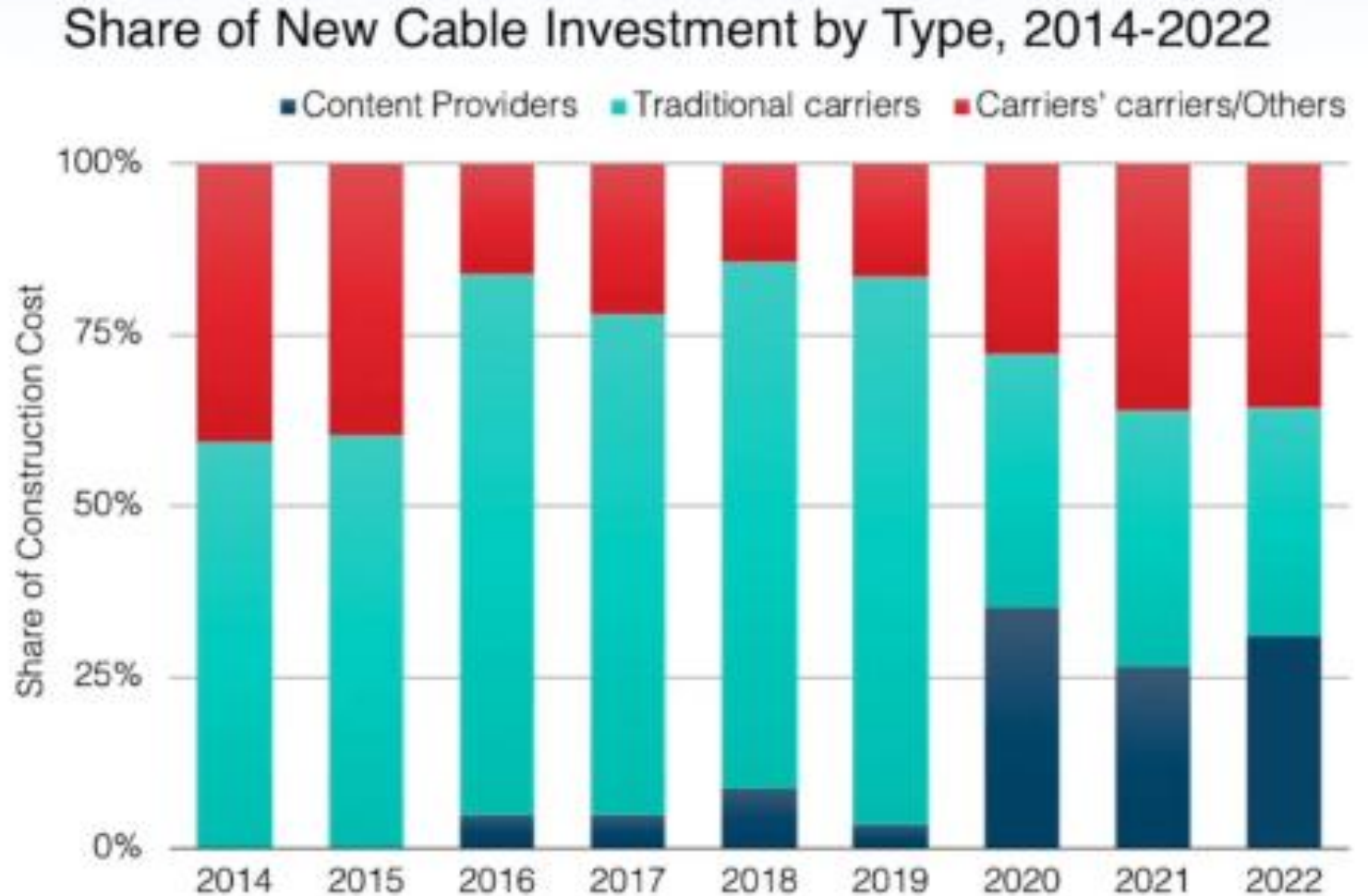
- IXPs and NSPs
- OTTs and CDNs

Cable Infrastructure

Traditional carriers
reduce investment

Content providers
increasingly invest
in fiber

Focus: subsea cables



telegeography.com

Recent Trends

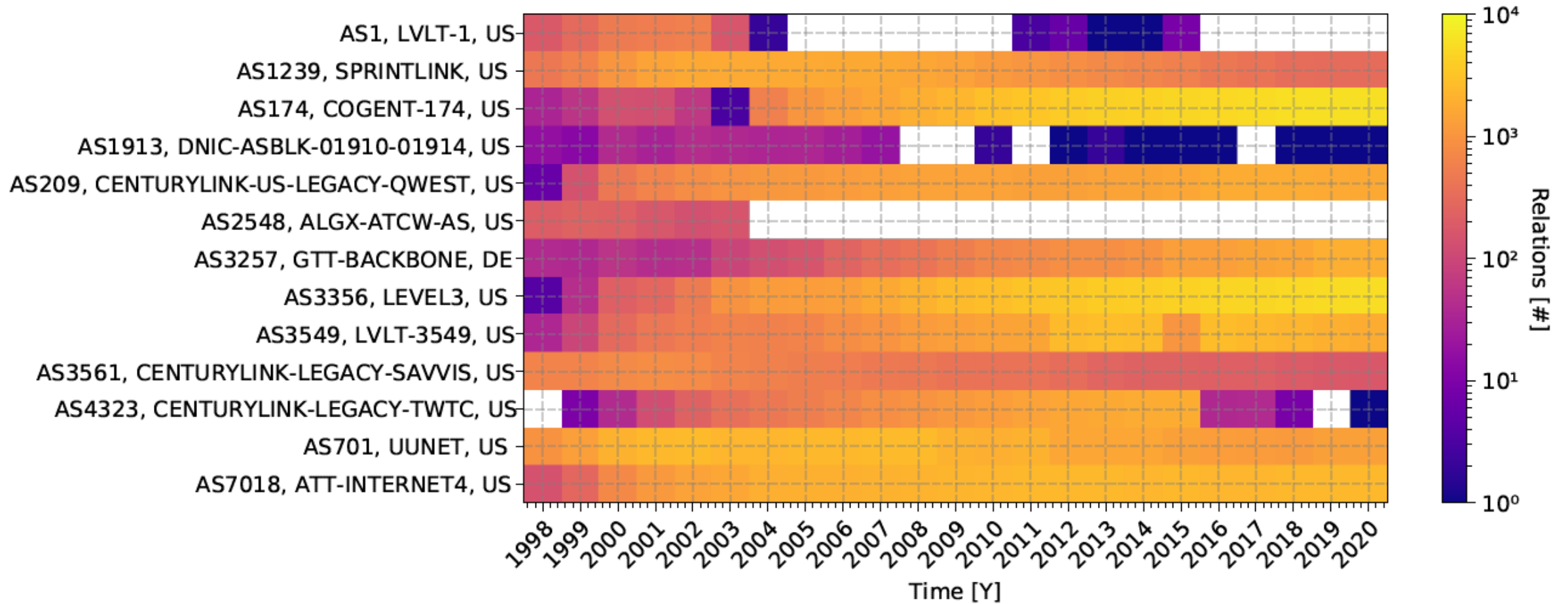
Reduce CAPEX in physical infrastructure

Merge with other Tier-1 or large IXPs

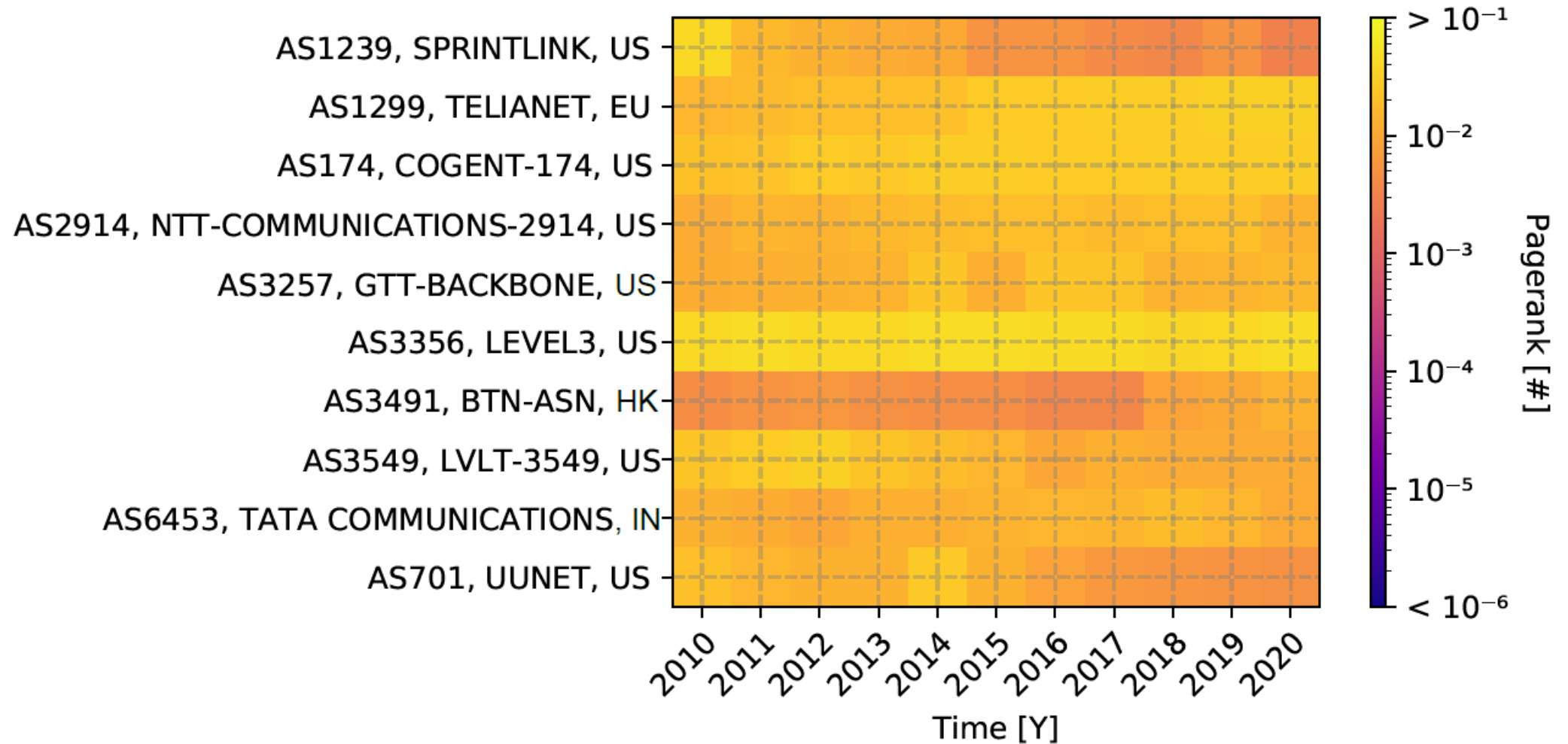
Increase presence in national access, enlarge consumer base

USP: Exclusive customer access

Customer Cones: Accumulated Top 5 ASes



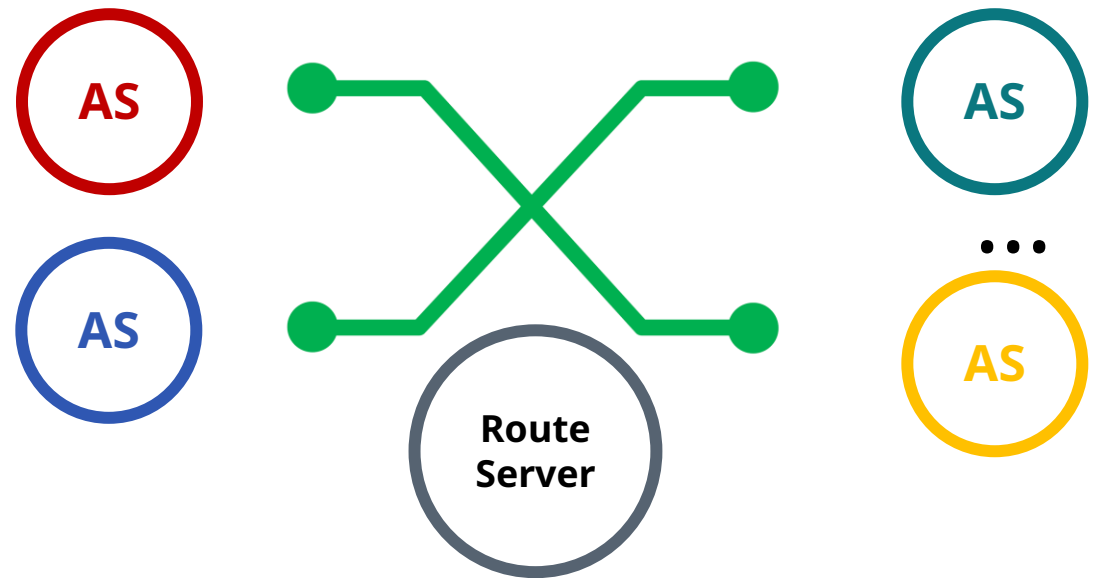
C2P Pagerank: Accumulated Top 5 ASes



Internet Exchange Points (IXPs)

IXPs exploit locality – they have highest gain, where neighbors shortcut transit

Utility of an IXP is larger the closer to consumers

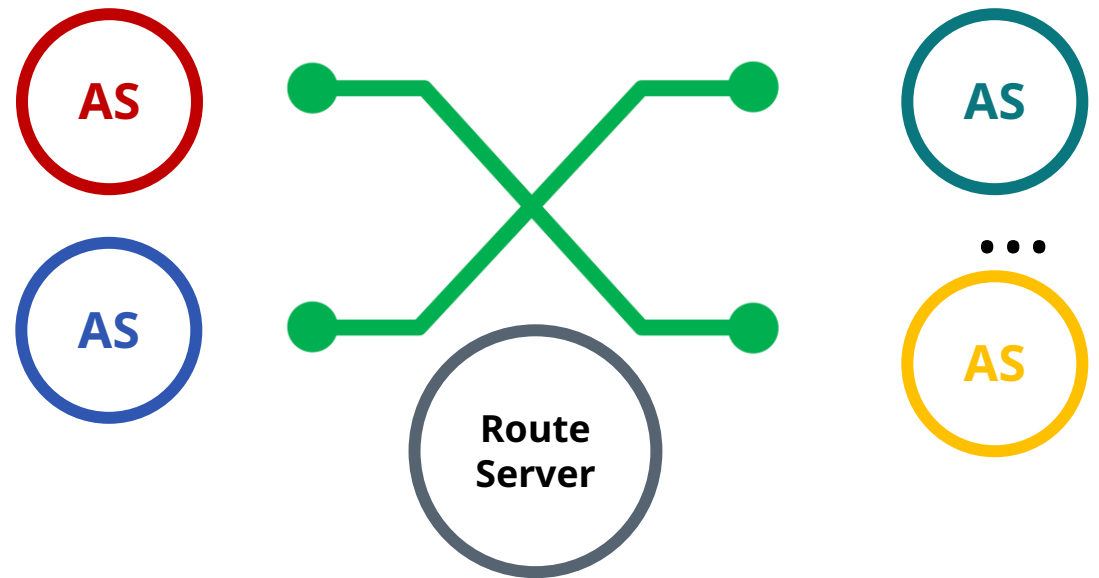


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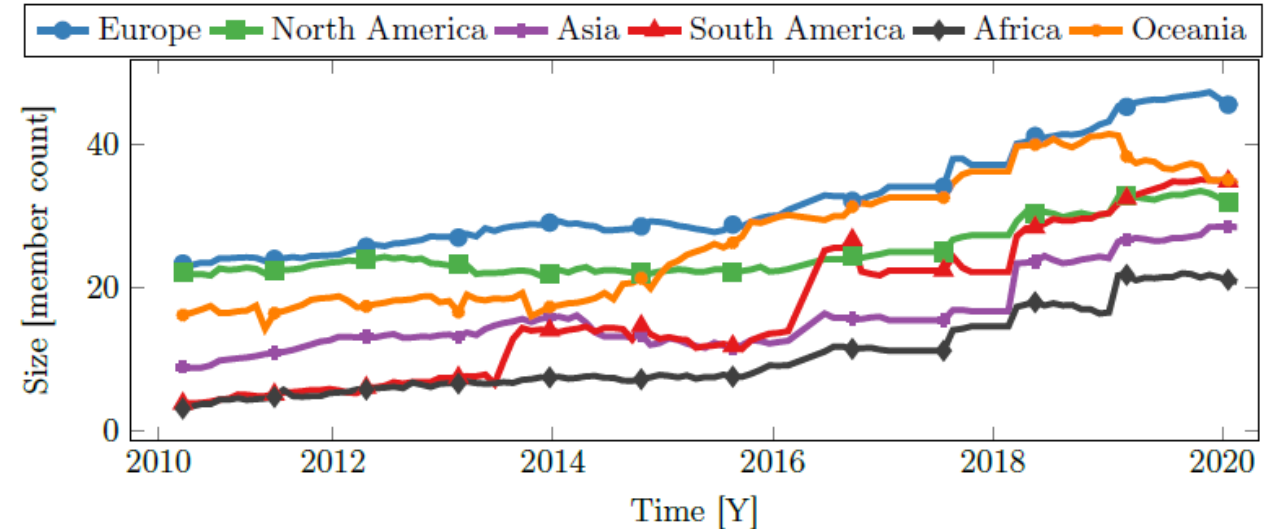
Utility of an IXP is larger the closer to consumers

“A cheap IX is probably a successful one.
An expensive IX is always a failure.”
-- Bill Woodcock, PCH

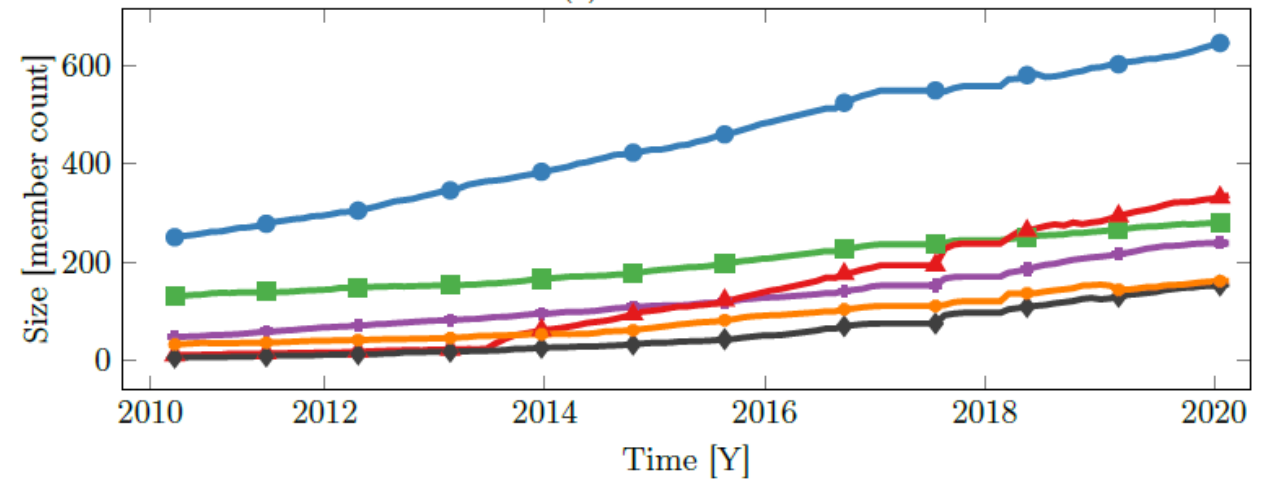


Global IXP Growth

In Europe, a small number of very large IXPs has emerged, which grow well above average



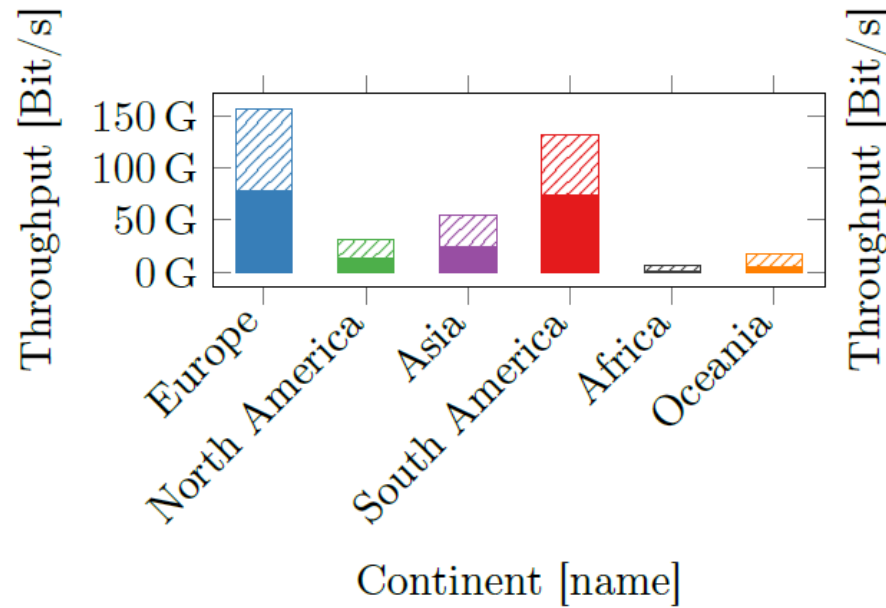
(a) All IXPs



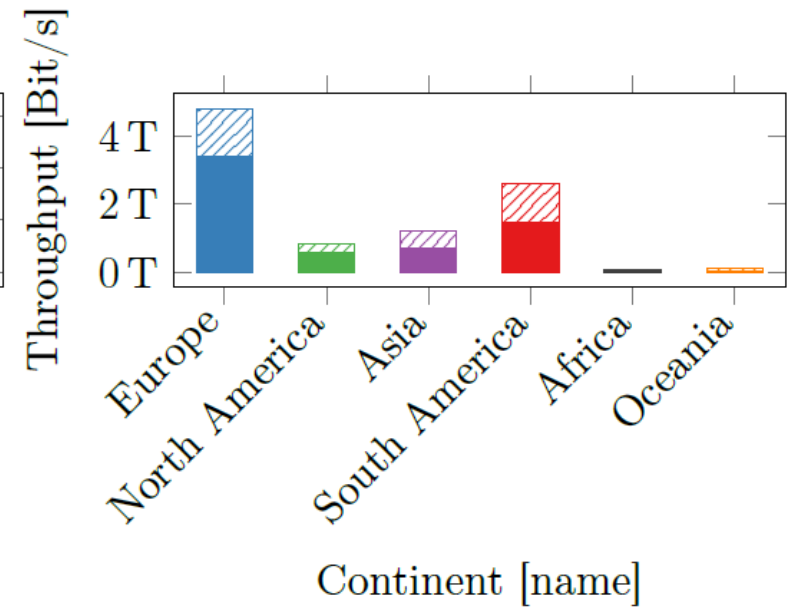
(b) Top five IXPs

Global IXP Throughput

Europe and South America (Brazil) take a different scale than the other continents

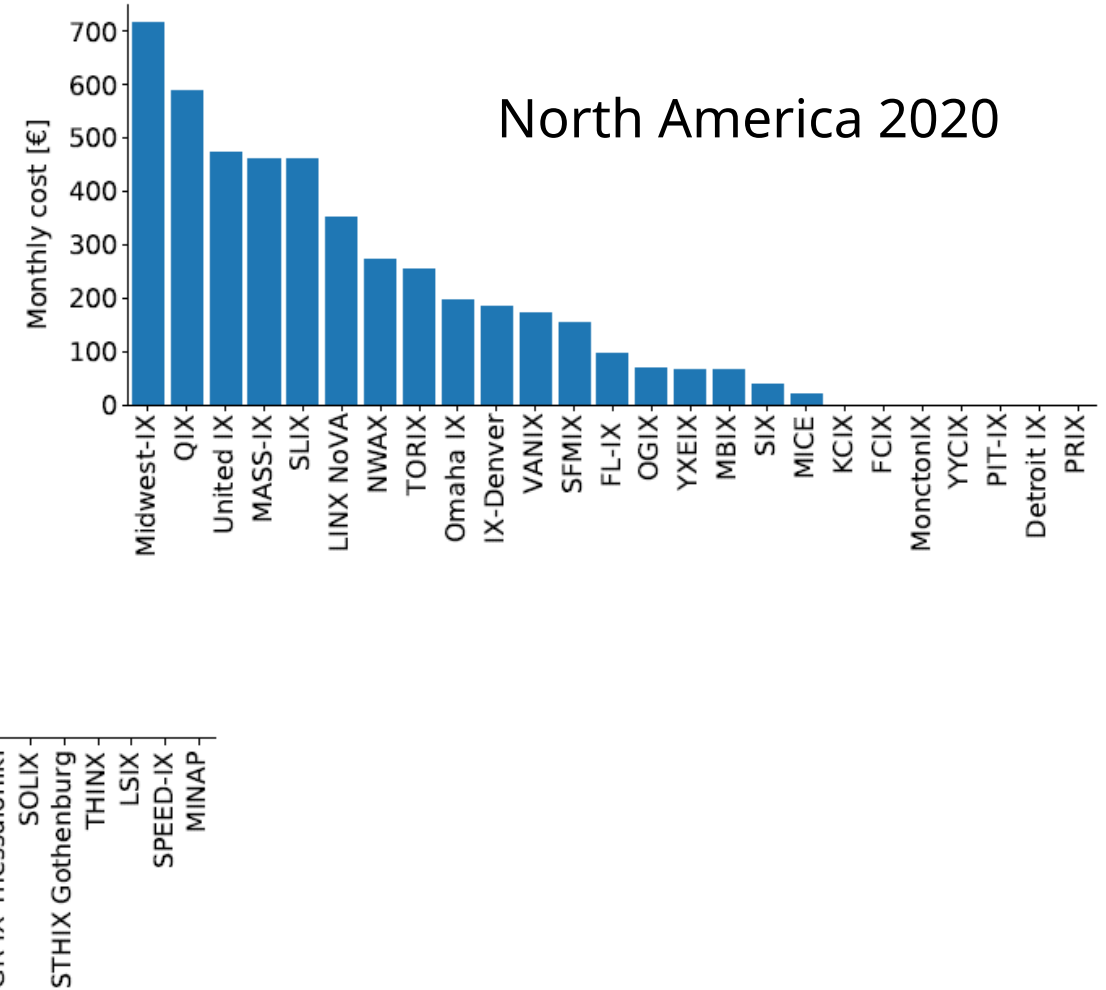
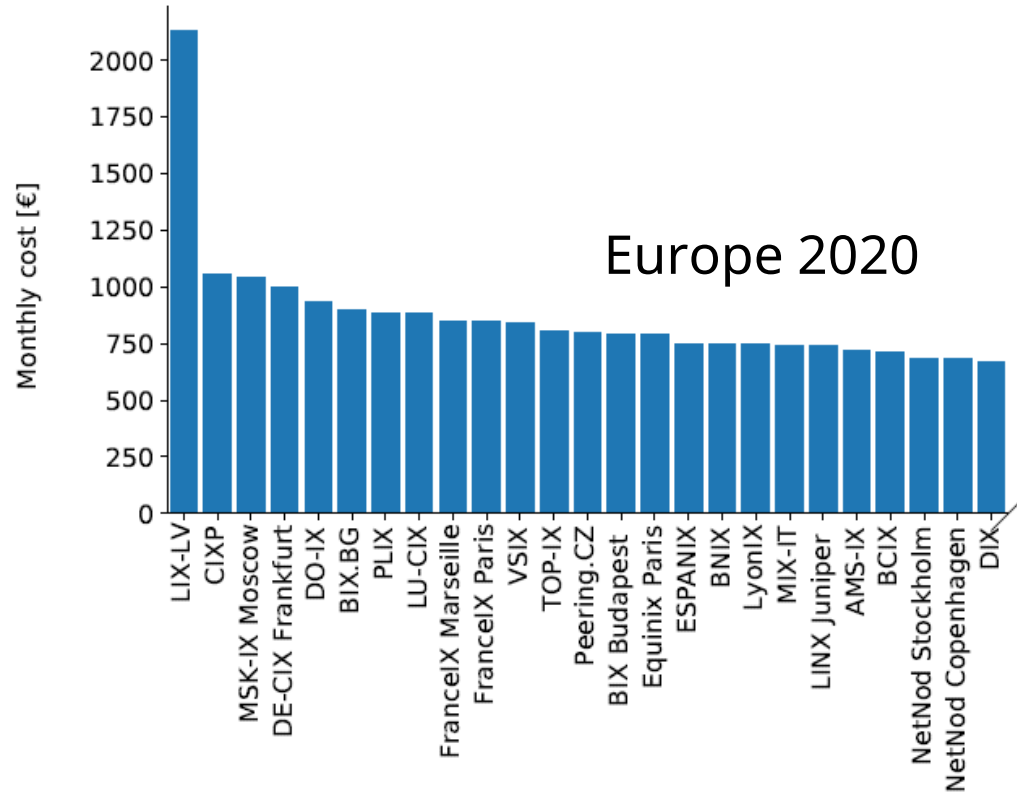


(a) All IXPs



(b) Top five IXPs by throughput

Global IXP Costs



OTT: Over-the-top Content providers



OTTs are popular because they don't rely on proprietary software



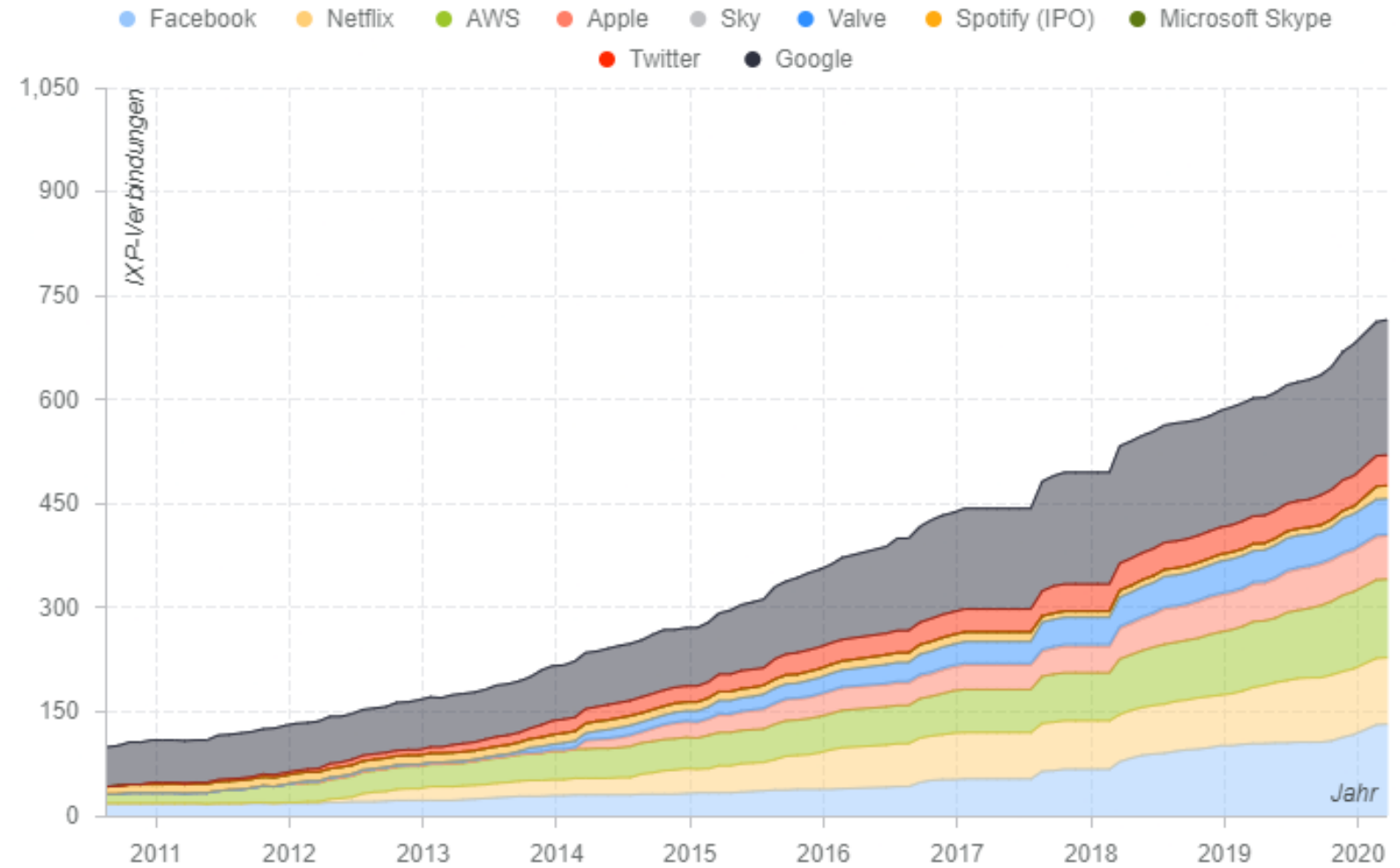
OTTs compete(d) with add-on services provided by ISPs (e.g., IPTV)



OTTs push for higher data rates (Ultra HD etc.)

OTTs operate their own CDN or use third-party CDNs.

Development of IXP connections for large OTTs



Two case OTT studies: Disney+ and Netflix

The Netflix logo is displayed in a bold, red, sans-serif font.

Originally using Akamai and Limelight
Since 2012 operating own CDN
Uses AWS to store and code movies



Two case OTT studies: Disney+ and Netflix

The Netflix logo is displayed in a bold, red, sans-serif font.

Originally using Akamai and Limelight
Since 2012 operating own CDN
Uses AWS to store and code movies



Started in USA in November 2019
Distributes Disney movies exclusively
No own CDN but Akamai, Lumen, Limelight,
Edgecast, CloudFront, and Fastly
Did not change the global data volume or peering

Two case OTT studies: Disney+ and Netflix

The Netflix logo, consisting of the word "NETFLIX" in a bold, red, sans-serif font.

Fragmentation for end users increases – an example where de-consolidation is not beneficial.

Network access

The Problem of Coverage & Competition

Provisioning of network access infrastructure is a challenging business outside metropolitan areas

Traditionally, public telephone monopolies provisioned basic network services – those were split up in the western countries

Network infrastructure w/ last mile coverage remains monolithic

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Network infrastructure w/ last mile coverage remains monolithic

How to organize a provider market that

- continuously invests into state-of-the-art access technologies?
- maintains and develops network coverage also in rural areas?
- allows for plurality in the last mile without replicating infrastructure?

The US Case

The US telephone monopolist AT&T was split up in 1984

- Seven independent regional Bell Operating Companies (Baby Bells)
- AT&T remained as long distance telephone company

➔ Geographic split w/o competition at consumers

Since then

- Southwestern Bell bought three other Baby Bells and later AT&T
- Atlantic Bell bought the remainders and formed Verizon

➔ Two large companies monopolize area-wise most of the US

Today, 50M households (40 %) only have a single provider choice

The German Case

Deutsche Telekom (DTAG) lost the network monopoly in 1996

Per law, the access to cable infrastructure was regulated

- DTAG kept its cables, but had to open access at regulated prices
- DTAG had to sell the TV cable network (CATV)

➔ Horizontal split across all last miles, competition at consumers

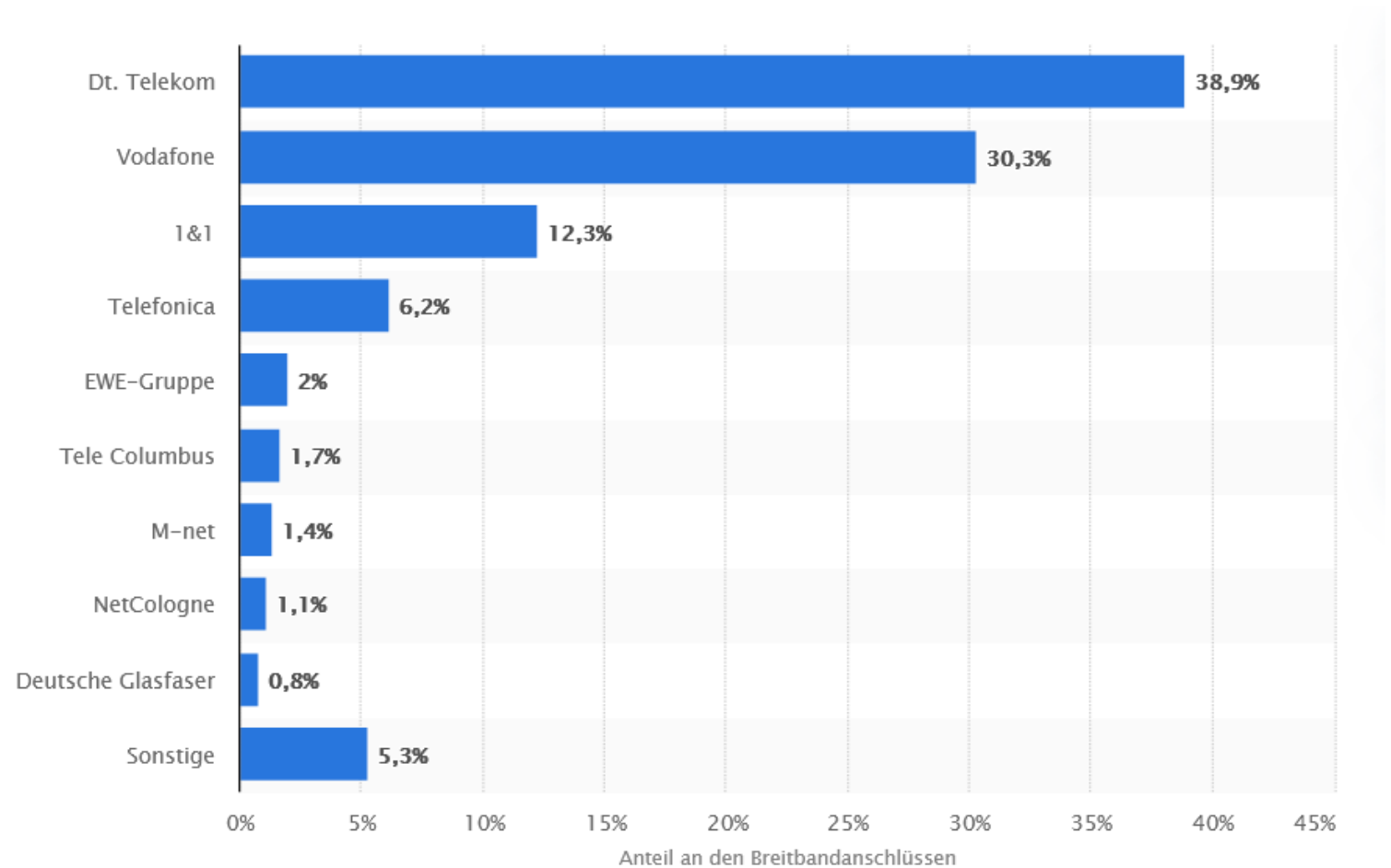
Since then

- Pluralistic network access at nation-wide prices
- TV cable network partly monopolized with Vodafone

➔ Diverse ecosystem of (partly regional) providers, relevant newcomers

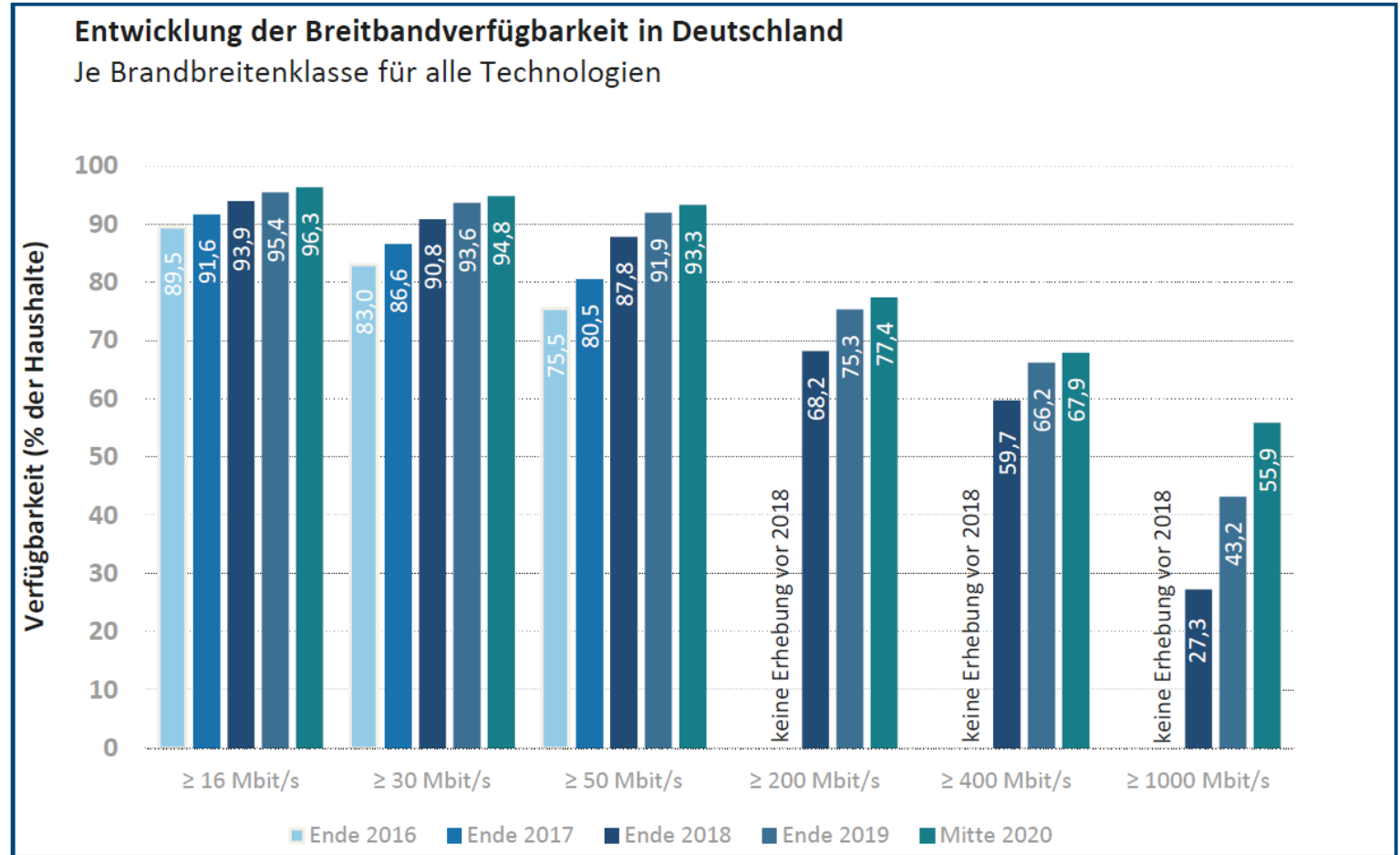
Broadband Access: German Market Shares (2020)

Source: Statista



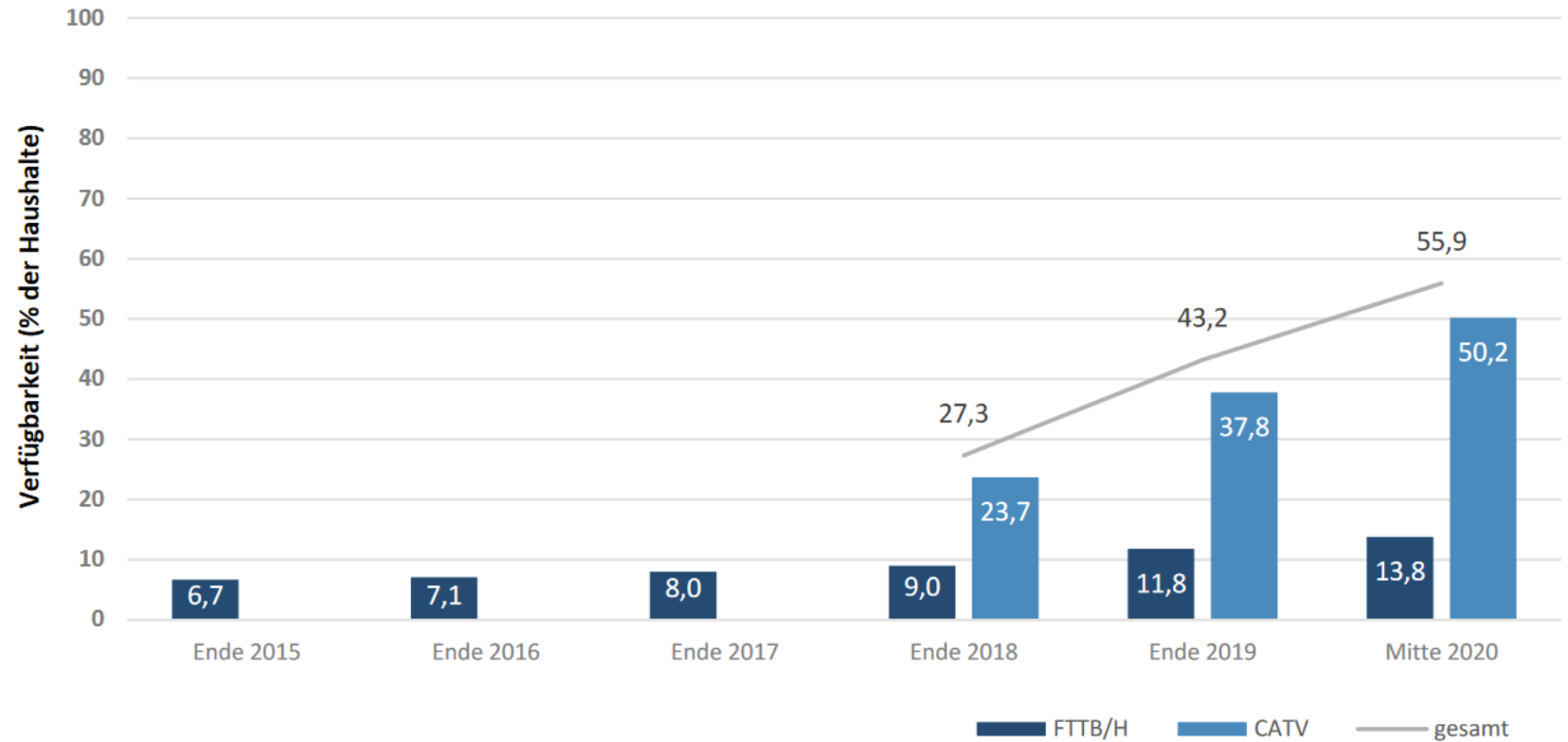
Broadband Access

Source: Breitbandatlas



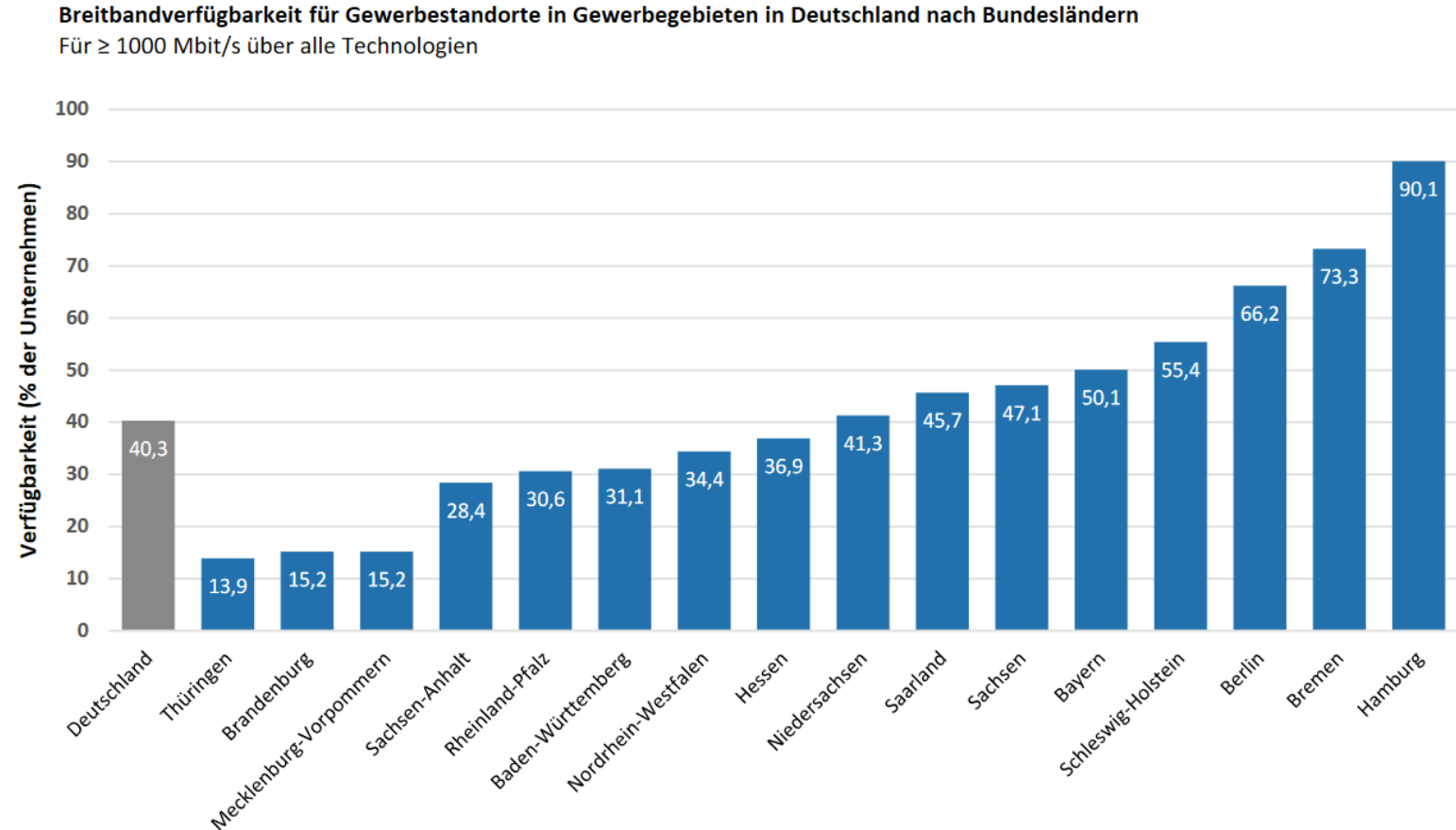
Gigabit Access

Source: Breitbandatlas



Gigabit Access in Industrial Zones

Source: Breitbandatlas



Internet applications

Content Delivery Networks (CDN)

Cloud Infrastructure Platform Services (CIPS)

CDNs and CIPS manifest the concept of global centralization

- Akamai dominates revenue
- Google and Tencent dominate growth
- Emerging competition: CIPS for IoT – currently dominated by Amazon

In 2019, 43% of the global Internet traffic was delivered by Google, Netflix, Facebook, Microsoft, Apple, or Amazon

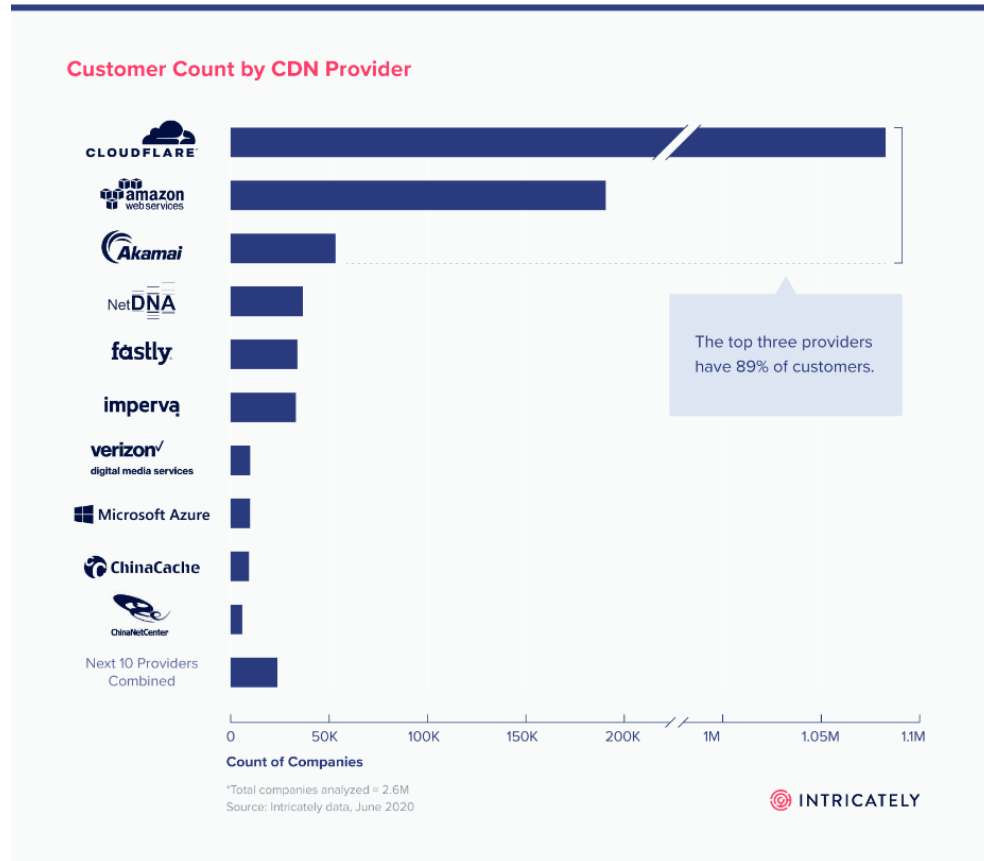
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End Systems

Operating Systems hold a key role in shaping the user interaction with the Internet

- First realized by Microsoft (late 90es) to kill Netscape
- Sophisticated by Apple's App store 2007
- Rebuilt by Google: Android, Google Play, Chrome ... Gmail, Quic, DNS, CT, ...

CAVEAT: The (pseudo-)HAL and its APIs of the OS may enforce undesired network use

Proprietary Applications and APIs

The growing use of APIs puts more of the Internet innovation, functionality, and interoperability into the hands of the dominant Internet platforms, whose interests may not always align with those of the broader technical community and other players.

- Internet Society

Digital Gatekeepers:
platforms to dominate markets

Social Networks & Media:
platforms to tie in users

Proprietary Communication Services:
platforms to monopolize user data

APIs owned by end systems or application
platforms abstract open Internet standards away

Why Apps Instead of the Web?

An app is just a website wrapped in enough intellectual property (IP) so that the company that made it can send you to prison if you dare to modify it so that it serves your interests rather than theirs.

- Cory Doctorow

Cory Doctorow: **Enshittification**, Verso Books, 2025

Apps push back the public Web, even though they are

- technologically reactionary
- more expensive to distribute and maintain

The Web is open, apps are closed

Apps are protected against reverse-engineering:

- US: Digital Millennium Copyright Act
- EU: Digital Copyright Act

Apps can successfully hide actions and intend

Social impact

Internet Access as Basic Service

Continuous access to the Internet has become essential in many areas of our lives:

- Information, education, social participation
- Various commercial & professional activities
- eGovernment & international exchange

Broadband Internet access today is limited in many countries, rural areas, and by social, economical, and age factors: Lack of options, knowledge, or interest leads to a 'digital illiteracy' in parts of our world.

Society faces a Digital Divide with increasing tension

Digital Divide

Children in urban areas are exposed to the digital world almost from the time they are born. However, children in tribal, rural communities may never even have access to primary education, let alone the digital world.

- Amit Chakravarty, ICRISAT

A significant number of people have no access to the Internet. Main reasons are:

- Lack of provider access
- Service cost
- Lack of end systems

Majority without Internet are children and young adults in rural areas

- Strongly correlated with educational deficits
- Excludes from public participation
- Leads to social marginalization

Economical, Political, and Social Divide

Broadband access opens business potentials

- Increased access boosts economy of rural places and neighboring regions

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Exclusion from Internet information reduces continuous education and political reflection

- Brexit voting in Britain showed this pattern

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Successful careers most often require broadband Internet, leaving undersupplied regions behind:

- Separating the elderly and the deprived

Safety and Disaster Management in an IP World

Emergency calls and safety alerting are integral, protective contributions of a country

- Traditional infrastructure was dedicated and vertically integrated
- Today, this is replaced by a horizontally structured all-IP-world

The horizontal, pluralistic IP system makes vertical infrastructure assurances difficult

New standards are needed to enable an integrated emergency management

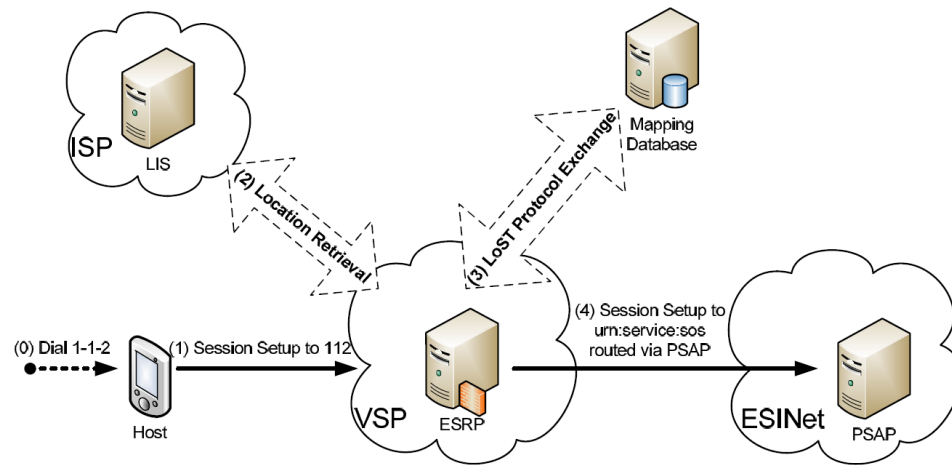
Internet Architecture for Emergency Calls

An incoming emergency call requires

- Locating the caller
- Mapping location to the responsible first responder
- Setting up the call with the local authority

All without active assistance
of the emergency caller

Internet Architecture for Emergency Calls



Source: Hannes Tschofening

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Security of Alerting Authorities

A horizontally integrated, pluralistic system requires effective security measures on each layer

This calls for thought- and careful operative procedures

Unfortunately, reality does not reflect these demands: alerting authorities are run as sloppy as any Internet service

DNS		Certificate		Assurance profile ¹	# Names
Restricted delegation	Supports DNSSEC	DV	O/EV		
✓	✓	–	✓	●	29 (≈ 2%)
✓	✓	✓	✗	◐	11
✗	✓	–	✓	◐	2
✓	✗	–	✓	◐	132
✗	✗	–	✓	◐	117
Total:					262 (≈ 20%)
✓	✗	✓	✗	○	354
✗	✗	✓	✗	○	482
✗	✓	✓	✗	○	3
✓	✓	✗	✗	○	2
✓	✗	✗	✗	○	67
✗	✓	✗	✗	○	2
✗	✗	✗	✗	○	126
Total:					1036 (≈ 78%)
Grand Total:					1327

¹ ● strong, ◐ weak, ○ inadequate (see Table 1)

Analysis of Alerting Authorities

Pouyan Fotouhi Tehrani, Eric Osterweil, J. Schiller, T. C. Schmidt, M. Wählisch,
Security of Alerting Authorities in the WWW: Measuring Namespaces, DNSSEC, and Web PKI, In: 30th The Web Conference (WWW'21), p. 2709–2720, ACM : New York, USA, April 2021.

Security of Alerting Authorities in the WWW: Measuring Namespaces, DNSSEC, and Web PKI

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ABSTRACT

During disasters, crisis, and emergencies the public relies on online services provided by official authorities to receive timely alerts, trustworthy information, and access to relief programs. It is therefore crucial for the authorities to reduce risks when accessing their online services. This includes catering to secure identification of service, secure resolution of name to network service, and content security and privacy as a minimum base for trustworthy communication.

In this paper, we take a first look at *Alerting Authorities* (AA) in the US and investigate security measures related to trustworthy and secure communication. We study the domain namespace structure, DNSSEC penetration, and web certificates. We introduce an integrative threat model to better understand whether and how the online presence and services of AAs are harmed. As an illustrative example, we investigate 1,388 Alerting Authorities. We observe partial heightened security relative to the global Internet trends, yet find cause for concern as about 78% of service providers fail to deploy

KEYWORDS

DNS, DNSSEC, Web PKI, Emergency Management

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1 INTRODUCTION

Online media have been proven to be an effective channel to communicate with the public. An ever growing number of Americans prefer to get their news online [41], social media is being used for public health announcements [87], and authorities provide public disaster education and services via Web portals [33]—just to mention a few examples. Communication of critical information such as emergency response [10, Chapter 3] and provisioning of

A Comprehensive Web Security Model

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A Security Model for Web-Based Communication

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Abstract

Web access involves various protocols to resolve domain names to IP addresses, establish data exchange channels with Web servers, and to authenticate communication partners. Each protocol has its own set of requirements and security measures. In addition to technical features, operating the Web also introduces organizational and political aspects which are important to consider when deploying a secure basis for Web-based communication.

In this paper, we propose an algorithmic security model based on the widely deployed technologies DNSSEC and Web PKI to cover three dimensions: *identification*, *resolution*, and *transaction*. Our model enables quantification and qualification of the security assurance provided by an online service provider. To verify the applicability of our model, we investigate the online presence of *Alerting Authorities* in the U.S., selected *German Emergency Service* providers, and *UN member states*. We observe partially enhanced security relative to global Internet trends, yet find cause for concern as only about 6% of unique hosts cater to secure resolution. About 46% of investigated organizations use shared certificates with 1% of all organizations having no or invalid certificates. Two thirds of organizations are

trast, the health department of Jackson County in Michigan, reachable under `www.co.jackson.mi.us`, is under a restricted top-level domain (TLD), that is, registration requires an eligible entity, which indicates that it belongs to Jackson County (`co.jackson`) in Michigan (`mi.us`). It is accompanied by an extended validation (EV) certificate, which serves as a definitive proof of identity.

In this paper, we introduce a generic security model for Web services based on three dimensions: resolution, transaction, and identification. Using this model, we provide a framework to understand whether and how specific integration of a service provider in the domain namespace and its use of DNSSEC and X.509 certificates impact secure communication (see Figure 1). Our model allows us to calculate an *Assurance Score* for a given service provider and qualify its level of security as an inadequate, a weak, or a strong *Assurance Profile*. We apply our model in practice by systematically investigating the Web representations of organizations active in emergency management or diplomatic relations. The starting points for our study are three datasets: (i) the list of *Alerting Authorities* (AA) in the U.S. (US-AA); (ii) list of fire departments, police, municipalities, and Red Cross chapters in the 200 most populous cities in Germany (DE-EM); and (iii) a list of

Content Consolidation: The Rabbit Hole

Content platforms aim at maximizing user participation time

- Use recommender systems to keep people interested
- Use larger content jumps to avoid boredom
- Use reinforcement learning to maximize excitement

This is reported to cause severe lock-ins of users – into a 'Rabbit Hole'

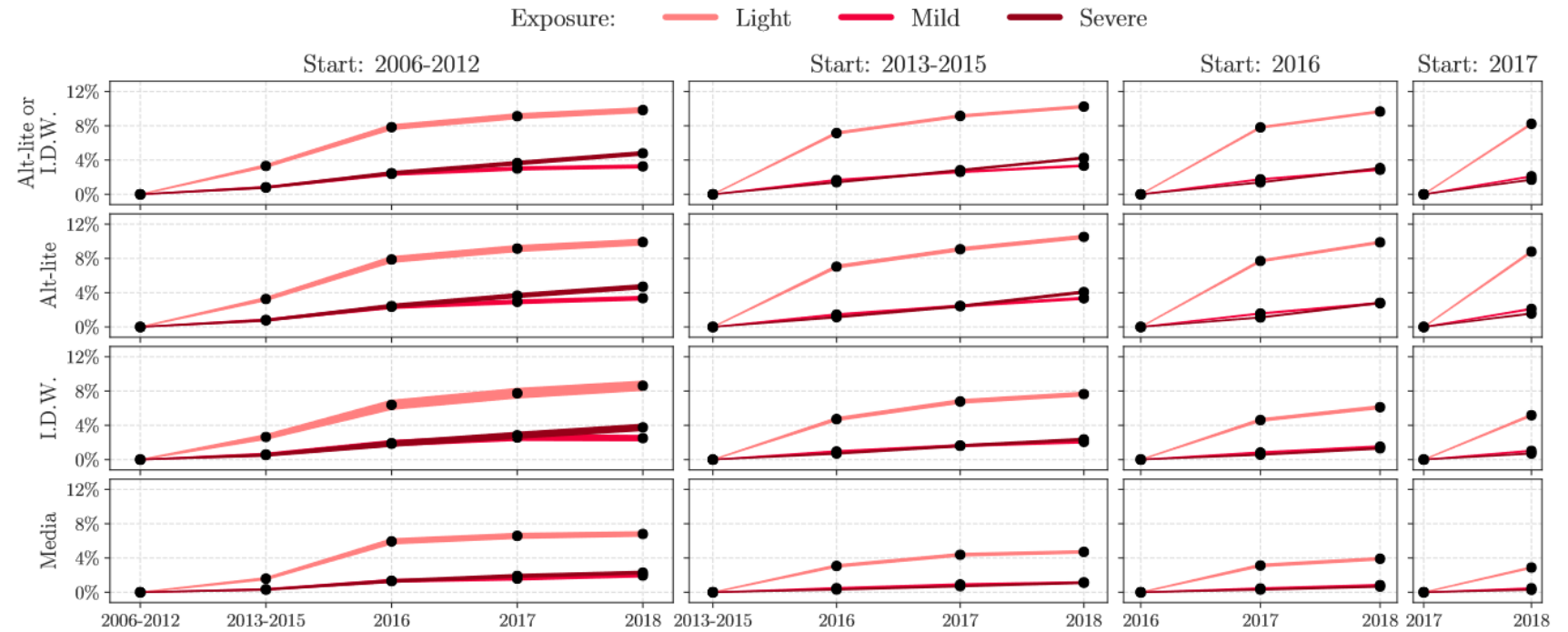
Recommender Impact

Transitions of different content consumers toward alt-right content after the first contact with this content.

Differentiated between intensity levels:

Light (1-2 comments),
Mild (3-5 comments);
Severe (6++ comments).

Effectiveness of recommendation system seems established in 2016.



Network Neutrality

Principle that Internet service providers (ISPs) must treat all Internet communications equally

Non-discriminatory forwarding of traffic, no blocking or slowing down, no specific charging

Regulatory issue:

US: “Save the Internet Act” (pending)

EU: “Regulation for laying down measures concerning open internet access and retail charges for regulated intra-EU communications” (2015/2020)

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